Derivation of the Debiased Lasso

A Step-by-Step Walkthrough

Ruilong Chen

April 21, 2026

Abstract

This note derives the debiased Lasso estimator of Zhang & Zhang [2014], van de Geer et al. [2014], and Javanmard & Montanari [2014] in five steps. We start from the Lasso KKT conditions, motivate the bias-correction term, show that the correction eliminates the first-order bias, establish asymptotic normality, and finally describe how the precision matrix is estimated via the nodewise Lasso of Meinshausen & Bühlmann [2006]. Throughout, we emphasize the role of predictor correlation, which controls the stability of every step in the construction.

1 Setup and Notation

Consider the high-dimensional linear model

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{X} \in \mathbb{R}^{n\times p}, \qquad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \boldsymbol{I}_n), \tag{1}$$

with $p \geq n$ and $\boldsymbol{\beta} \in \mathbb{R}^p$ assumed to be sparse, with support $S = \{j : \beta_j \neq 0\}$ and sparsity level $s = |S|$. We denote the empirical Gram matrix by

$$\hat{\boldsymbol{\Sigma}} = \frac{1}{n}\boldsymbol{X}^\top \boldsymbol{X},$$

and write $\boldsymbol{\Sigma} = \mathbb{E}[\hat{\boldsymbol{\Sigma}}]$, with population precision matrix $\boldsymbol{\Theta} = \boldsymbol{\Sigma}^{-1}$.

2 Step 1: Lasso and Its KKT Conditions

The Lasso estimator [Tibshirani, 1996] is defined as

$$\hat{\boldsymbol{\beta}}^{\text{Lasso}} = \arg\min_{\boldsymbol{\beta}\in\mathbb{R}^p} \left\{ \frac{1}{2n}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_1 \right\}. \tag{2}$$

The objective is convex but non-smooth at $\beta_j = 0$. We therefore characterize the optimum using subgradients.

The subdifferential of $|\beta_j|$ is

$$\partial|\beta_j| = \begin{cases} \{+1\} & \beta_j > 0, \\ \{-1\} & \beta_j < 0, \\ [-1, +1] & \beta_j = 0. \end{cases}$$

The first-order optimality (KKT) condition for (2) states that there exists $\hat{\boldsymbol{\kappa}} \in \mathbb{R}^p$ with $\hat{\boldsymbol{\kappa}}_j \in \partial|\hat{\beta}_j^{\text{Lasso}}|$ such that

$$\boxed{\frac{1}{n}\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{\text{Lasso}}) = \lambda\hat{\boldsymbol{\kappa}}.} \tag{3}$$

Interpretation. For OLS, the analogous condition is $\frac{1}{n}\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{\text{OLS}}) = \mathbf{0}$: the residual is orthogonal to every column of $\boldsymbol{X}$. For Lasso, the residual is *not* orthogonal — it is offset by $\lambda\hat{\boldsymbol{\kappa}}$. This nonzero offset is precisely the source of the regularization bias, and it is exactly what the debiased estimator will correct for.

3 Step 2: Constructing the Bias-Correction Term

We want an estimator $\tilde{\beta}_j$ that is approximately unbiased and has a tractable asymptotic distribution. Starting from the KKT condition (3), substitute the model $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$:

$$\begin{aligned}\frac{1}{n}\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{\text{Lasso}}) &= \frac{1}{n}\boldsymbol{X}^\top\boldsymbol{X}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}^{\text{Lasso}}) + \frac{1}{n}\boldsymbol{X}^\top\boldsymbol{\varepsilon} \\ &= \hat{\boldsymbol{\Sigma}}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}^{\text{Lasso}}) + \frac{1}{n}\boldsymbol{X}^\top\boldsymbol{\varepsilon}. \end{aligned} \tag{4}$$

The low-dimensional intuition. If $\hat{\boldsymbol{\Sigma}}$ were invertible, we could solve (4) for the estimation error:

$$\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}^{\text{Lasso}} = \hat{\boldsymbol{\Sigma}}^{-1} \cdot \frac{1}{n}\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{\text{Lasso}}) - \hat{\boldsymbol{\Sigma}}^{-1} \cdot \frac{1}{n}\boldsymbol{X}^\top\boldsymbol{\varepsilon}.$$

This would give us $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}^{\text{Lasso}} + \hat{\boldsymbol{\Sigma}}^{-1} \cdot \frac{1}{n}\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{\text{Lasso}}) - (\text{noise})$, which is essentially the OLS solution applied to the Lasso residual. The non-noise part of the right-hand side could serve as our debiased estimator.

The high-dimensional obstacle. When $p > n$, $\hat{\boldsymbol{\Sigma}}$ is singular and cannot be inverted. We therefore replace $\hat{\boldsymbol{\Sigma}}^{-1}$ by an *approximate inverse* $\hat{\boldsymbol{\Theta}}$, an estimator of the precision matrix $\boldsymbol{\Theta} = \boldsymbol{\Sigma}^{-1}$. The construction of $\hat{\boldsymbol{\Theta}}$ is deferred to Section 6.

The debiased Lasso. Using the j-th row $\hat{\boldsymbol{\Theta}}_j$ of $\hat{\boldsymbol{\Theta}}$, define

$$\boxed{\tilde{\beta}_j \;=\; \hat{\beta}_j^{\text{Lasso}} + \frac{1}{n}\hat{\boldsymbol{\Theta}}_j\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{\text{Lasso}}).} \tag{5}$$

This is the debiased (or *desparsified*) Lasso [Zhang & Zhang, 2014, van de Geer et al., 2014, Javanmard & Montanari, 2014]. The correction term is exactly the projection of the Lasso residual onto direction j of the parameter space, weighted by $\hat{\boldsymbol{\Theta}}_j$.

4 Step 3: Why the Correction Eliminates the Bias

We now show that the leading bias in $\tilde{\beta}_j$ vanishes. Substituting (4) into (5):

$$\begin{aligned}\tilde{\beta}_j &= \hat{\beta}_j^{\text{Lasso}} + \frac{1}{n}\hat{\boldsymbol{\Theta}}_j\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{\text{Lasso}}) \\ &= \hat{\beta}_j^{\text{Lasso}} + \hat{\boldsymbol{\Theta}}_j\hat{\boldsymbol{\Sigma}}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}^{\text{Lasso}}) + \frac{1}{n}\hat{\boldsymbol{\Theta}}_j\boldsymbol{X}^\top\boldsymbol{\varepsilon}. \end{aligned} \tag{6}$$

Subtract β_j from both sides:

$$\begin{aligned}
\tilde{\beta}_j - \beta_j &= (\hat{\beta}_j^{\text{Lasso}} - \beta_j) + \hat{\boldsymbol{\Theta}}_j \hat{\boldsymbol{\Sigma}}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}^{\text{Lasso}}) + \frac{1}{n}\hat{\boldsymbol{\Theta}}_j \boldsymbol{X}^\top \boldsymbol{\varepsilon} \\
&= -\boldsymbol{e}_j^\top (\hat{\boldsymbol{\beta}}^{\text{Lasso}} - \boldsymbol{\beta}) + \hat{\boldsymbol{\Theta}}_j \hat{\boldsymbol{\Sigma}}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}^{\text{Lasso}}) + \frac{1}{n}\hat{\boldsymbol{\Theta}}_j \boldsymbol{X}^\top \boldsymbol{\varepsilon} \\
&= -\big(\boldsymbol{e}_j^\top - \hat{\boldsymbol{\Theta}}_j \hat{\boldsymbol{\Sigma}}\big)(\hat{\boldsymbol{\beta}}^{\text{Lasso}} - \boldsymbol{\beta}) + \frac{1}{n}\hat{\boldsymbol{\Theta}}_j \boldsymbol{X}^\top \boldsymbol{\varepsilon}, \qquad (7)
\end{aligned}$$

where $\boldsymbol{e}_j$ is the j-th standard basis vector. We have arrived at the fundamental decomposition of the debiased Lasso:

$$\boxed{\sqrt{n}(\tilde{\beta}_j - \beta_j) \;=\; \underbrace{\frac{1}{\sqrt{n}}\hat{\boldsymbol{\Theta}}_j \boldsymbol{X}^\top \boldsymbol{\varepsilon}}_{\text{noise term } Z_j} \;-\; \underbrace{\sqrt{n}\big(\hat{\boldsymbol{\Theta}}_j \hat{\boldsymbol{\Sigma}} - \boldsymbol{e}_j^\top\big)(\hat{\boldsymbol{\beta}}^{\text{Lasso}} - \boldsymbol{\beta})}_{\text{remainder } \Delta_j}.} \qquad (8)$$

The bias is killed when $\hat{\boldsymbol{\Theta}}_j \hat{\boldsymbol{\Sigma}} \approx \boldsymbol{e}_j^\top$. If $\hat{\boldsymbol{\Theta}}$ were an exact inverse of $\hat{\boldsymbol{\Sigma}}$, then $\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{\Sigma}} = \boldsymbol{I}$ and the remainder Δ_j would vanish identically. In high dimensions $\hat{\boldsymbol{\Theta}}$ is only an *approximate* inverse, but a careful construction makes $\hat{\boldsymbol{\Theta}}_j \hat{\boldsymbol{\Sigma}} - \boldsymbol{e}_j^\top$ small in ℓ_∞-norm. By Hölder's inequality,

$$|\Delta_j| \le \sqrt{n}\, \|\hat{\boldsymbol{\Theta}}_j \hat{\boldsymbol{\Sigma}} - \boldsymbol{e}_j\|_\infty \cdot \|\hat{\boldsymbol{\beta}}^{\text{Lasso}} - \boldsymbol{\beta}\|_1.$$

Standard Lasso theory [Bickel et al., 2009] gives $\|\hat{\boldsymbol{\beta}}^{\text{Lasso}} - \boldsymbol{\beta}\|_1 = O_P(s\sqrt{\log p/n})$ under a restricted eigenvalue condition. The nodewise construction of $\hat{\boldsymbol{\Theta}}_j$ in Section 6 ensures $\|\hat{\boldsymbol{\Theta}}_j \hat{\boldsymbol{\Sigma}} - \boldsymbol{e}_j\|_\infty = O_P(\sqrt{\log p/n})$. Combining,

$$|\Delta_j| = O_P\left(\frac{s \log p}{\sqrt{n}}\right),$$

which is $o_P(1)$ as long as $s \log p = o(\sqrt{n})$. This is the standard *ultra-sparsity* condition required for asymptotic normality of the debiased estimator [van de Geer et al., 2014].

5 Step 4: Asymptotic Normality

Under the sparsity condition above, the remainder Δ_j is asymptotically negligible, and (8) reduces to

$$\sqrt{n}(\tilde{\beta}_j - \beta_j) = Z_j + o_P(1), \qquad Z_j := \frac{1}{\sqrt{n}}\hat{\boldsymbol{\Theta}}_j \boldsymbol{X}^\top \boldsymbol{\varepsilon}. \qquad (9)$$

The noise term Z_j is a linear combination of the i.i.d. noise variables $\varepsilon_1, \ldots, \varepsilon_n$:

$$Z_j = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} (\hat{\boldsymbol{\Theta}}_j \boldsymbol{X}^\top)_i \, \varepsilon_i.$$

Conditional on $\boldsymbol{X}$ (and hence on $\hat{\boldsymbol{\Theta}}$), this is a sum of independent mean-zero random variables. By the Lindeberg–Feller central limit theorem,

$$\boxed{\sqrt{n}(\tilde{\beta}_j - \beta_j) \;\xrightarrow{d}\; N\big(0,\; \sigma^2 \hat{\boldsymbol{\Theta}}_j \hat{\boldsymbol{\Sigma}} \hat{\boldsymbol{\Theta}}_j^\top\big).} \qquad (10)$$

The asymptotic variance can be consistently estimated by

$$\hat{\tau}_j^2 = \hat{\sigma}^2 \hat{\boldsymbol{\Theta}}_j \hat{\boldsymbol{\Sigma}} \hat{\boldsymbol{\Theta}}_j^\top,$$

where $\hat{\sigma}^2$ is a Lasso-based variance estimator (e.g., the scaled Lasso of Sun & Zhang 2012). A two-sided $(1-\alpha)$ confidence interval is then

$$\mathrm{CI}_{1-\alpha}(\beta_j) = \left[\tilde{\beta}_j - z_{1-\alpha/2} \frac{\hat{\tau}_j}{\sqrt{n}}, \;\; \tilde{\beta}_j + z_{1-\alpha/2} \frac{\hat{\tau}_j}{\sqrt{n}} \right]. \tag{11}$$

6 Step 5: Estimating $\hat{\boldsymbol{\Theta}}_j$ via the Nodewise Lasso

It remains to construct $\hat{\boldsymbol{\Theta}}_j$, an estimator of the j-th row of the precision matrix $\boldsymbol{\Theta} = \boldsymbol{\Sigma}^{-1}$. The approach of van de Geer et al. [2014], building on Meinshausen & Bühlmann [2006], is the *nodewise Lasso*.

Key identity. Consider regressing $\boldsymbol{X}_j$ on the remaining columns $\boldsymbol{X}_{-j}$:

$$\boldsymbol{X}_j = \boldsymbol{X}_{-j}\boldsymbol{\gamma}_j + \boldsymbol{\eta}_j, \qquad \boldsymbol{\eta}_j \perp\!\!\!\perp \boldsymbol{X}_{-j}.$$

Standard linear-algebra results identify

$$\boldsymbol{\gamma}_j = -\frac{\boldsymbol{\Theta}_{j,-j}}{\Theta_{jj}}, \qquad \mathrm{Var}(\eta_{ji}) = \frac{1}{\Theta_{jj}} =: \tau_j^2.$$

That is, the j-th row of $\boldsymbol{\Theta}$ is fully determined by the regression coefficients of $\boldsymbol{X}_j$ on $\boldsymbol{X}_{-j}$ and the corresponding residual variance.

The nodewise Lasso estimator. Estimate $\boldsymbol{\gamma}_j$ in the high-dimensional regime by Lasso:

$$\hat{\gamma}_j = \arg\min_{\boldsymbol{\gamma}\in\mathbb{R}^{p-1}} \left\{ \frac{1}{2n}\|\boldsymbol{X}_j - \boldsymbol{X}_{-j}\boldsymbol{\gamma}\|_2^2 + \mu_j\|\boldsymbol{\gamma}\|_1 \right\}, \tag{12}$$

and define the residual variance estimate

$$\hat{\tau}_j^2 = \frac{1}{n}\|\boldsymbol{X}_j - \boldsymbol{X}_{-j}\hat{\boldsymbol{\gamma}}_j\|_2^2 + \mu_j\|\hat{\boldsymbol{\gamma}}_j\|_1.$$

Then the j-th row of $\hat{\boldsymbol{\Theta}}$ is constructed as

$$\hat{\boldsymbol{\Theta}}_j = \frac{1}{\hat{\tau}_j^2}\big(-\hat{\gamma}_{j,1}, \; \ldots, \; -\hat{\gamma}_{j,j-1}, \; 1, \; -\hat{\gamma}_{j,j+1}, \; \ldots, \; -\hat{\gamma}_{j,p}\big). \tag{13}$$

By construction, this $\hat{\boldsymbol{\Theta}}_j$ approximately satisfies $\hat{\boldsymbol{\Theta}}_j\hat{\boldsymbol{\Sigma}} \approx \boldsymbol{e}_j^\top$, which is precisely what was required for the bias-correction step in Section 3.

Why correlation breaks this estimator. Suppose two columns $\boldsymbol{X}_j$ and $\boldsymbol{X}_k$ are highly correlated — for example, two SNPs in strong linkage disequilibrium. Then:

- Predicting $\boldsymbol{X}_j$ from $\boldsymbol{X}_{-j}$ is *easy* because $\boldsymbol{X}_k$ alone explains most of $\boldsymbol{X}_j$.
- But the Lasso regression (12) cannot reliably decide *which* of the correlated variables ($\boldsymbol{X}_k$ or other co-correlated columns) should carry the weight. The estimated $\hat{\boldsymbol{\gamma}}_j$ becomes unstable across samples.
- More importantly, $\hat{\tau}_j^2 \to 0$, since the residual is small.

Because $\hat{\mathbf{\Theta}}_j$ contains $1/\hat{\tau}_j^2$ as a multiplicative factor in (13), this factor blows up:

$$\|\hat{\mathbf{\Theta}}_j\|_2 = \frac{1}{\hat{\tau}_j^2} \cdot \sqrt{1 + \|\hat{\gamma}_j\|_2^2} \;\longrightarrow\; \infty.$$

Through the asymptotic variance formula (10), this directly inflates $\mathrm{Var}(\tilde{\beta}_j)$. The resulting confidence intervals become extremely wide — or numerically unstable — exactly when predictors are most correlated. This is the predicted failure mode of debiased Lasso under correlated designs and the motivation for the empirical comparison in this project.

7 Summary

1. **Lasso KKT.** The Lasso residual is not orthogonal to $\boldsymbol{X}$: it is offset by $\lambda\hat{\boldsymbol{\kappa}}$. This offset is the source of regularization bias.
2. **Correction.** Multiply the residual by an approximate inverse $\hat{\mathbf{\Theta}}_j$ of $\hat{\mathbf{\Sigma}}$ and add it to $\hat{\beta}_j^{\text{Lasso}}$.
3. **Bias removal.** The decomposition $\sqrt{n}(\tilde{\beta}_j - \beta_j) = Z_j - \Delta_j$ shows that bias is killed up to a remainder that vanishes under $s \log p = o(\sqrt{n})$.
4. **Asymptotic normality.** The dominant term Z_j is a CLT-amenable linear combination of i.i.d. noise, giving $\sqrt{n}(\tilde{\beta}_j - \beta_j) \to N(0, \sigma^2 \hat{\mathbf{\Theta}}_j \hat{\mathbf{\Sigma}} \hat{\mathbf{\Theta}}_j^\top)$.
5. **Precision matrix estimation.** $\hat{\mathbf{\Theta}}_j$ is constructed from a nodewise Lasso of $\boldsymbol{X}_j$ on $\boldsymbol{X}_{-j}$. Strong predictor correlation drives the residual variance toward zero, inflating $\hat{\mathbf{\Theta}}_j$ and ultimately the confidence intervals — the central reason debiased inference degrades under correlated designs.

References

Bickel, P. J., Ritov, Y., & Tsybakov, A. B. (2009). Simultaneous analysis of Lasso and Dantzig selector. *Annals of Statistics*, 37(4), 1705–1732.

Javanmard, A., & Montanari, A. (2014). Confidence intervals and hypothesis testing for high-dimensional regression. *Journal of Machine Learning Research*, 15, 2869–2909.

Meinshausen, N., & Bühlmann, P. (2006). High-dimensional graphs and variable selection with the Lasso. *Annals of Statistics*, 34(3), 1436–1462.

Sun, T., & Zhang, C.-H. (2012). Scaled sparse linear regression. *Biometrika*, 99(4), 879–898.

Tibshirani, R. (1996). Regression shrinkage and selection via the Lasso. *Journal of the Royal Statistical Society: Series B*, 58(1), 267–288.

van de Geer, S., Bühlmann, P., Ritov, Y., & Dezeure, R. (2014). On asymptotically optimal confidence regions and tests for high-dimensional models. *Annals of Statistics*, 42(3), 1166–1202.

Zhang, C.-H., & Zhang, S. S. (2014). Confidence intervals for low dimensional parameters in high dimensional linear models. *Journal of the Royal Statistical Society: Series B*, 76(1), 217–242.